Teacher notes – remove before printing

Heating water with an immersion heater – the relationship between heat and temperature change

Syllabus link: QCAA Physics 2025 General Senior Syllabus, Unit 1: Thermal, nuclear and electrical physics – heating processes (specific heat capacity, $Q = mc\Delta T$); electrical energy and power ($Q = VIt$). Also builds the data-analysis and evaluation skills needed for the IA1 Data test and IA2 Student experiment.

Preparation and equipment notes

- A 12 V, ~50 W heater draws ~4 A. Check the power supply can deliver at least 5 A continuously and set the ammeter to its 10 A range (a 2 A fused range will blow).
- Pre-test each heater. Heaters that have been run in air often have damaged elements – replace any that are discoloured or give erratic current.
- Between trials the heater is hot. Keep a beaker of cold water at each station so students can cool it (power off) before starting the next trial, otherwise trials 2 and 3 get a “head start” and the spread increases.
- Heating stops at 60 °C to limit scald risk and to keep evaporation and losses small. With 150 g and 50 W, most groups reach about 50–56 °C at 8 minutes from a ~22 °C start.
- If probes/data-loggers are used, students still record at 60 s intervals so the tables and Excel steps work unchanged.

Pre-lab answers

- **Independent:** heat added, Q (J) – varied by heating time, calculated with $Q = VIt$.
- **Dependent:** temperature change of the water, ΔT (°C).
- **Controlled:** mass of water (~150 g, balance); heater power (same supply setting, same heater); starting temperature (fresh room-temperature water); insulation (same cup and lid); stirring rate; thermometer position (clear of the element, same depth).
- **Predicted graph:** a straight line through the origin.
- **Reaction time:** human reaction time is ~0.2–0.3 s at both starting and reading the stopwatch, and the temperature must be read at the same instant as the time – so the real timing uncertainty is roughly 10–30 times larger than the 0.01 s resolution. (Relative to 60 s intervals this is still small, ~0.5%.)
- **Instrument uncertainties** (typical): balance ±0.01 g; alcohol thermometer with 1 °C divisions ±0.5 °C; digital probe ±0.1 °C; digital voltmeter ±0.01 V or ±0.1 V; digital ammeter ±0.01 A; stopwatch ±0.01 s.

Expected results with typical values

Typical data for m = 0.150 kg, mean V = 12.1 V, mean I = 4.15 A (P = 50.2 W):

Table T1 Typical results

t (s)	Trial 1 (°C)	Trial 2 (°C)	Trial 3 (°C)	Mean *T* (°C)	± ½ range (°C)	*Q* (J)	Δ*T* (°C)
0	22.0	21.5	22.5	22.0	0.5	0	0.0
60	25.0	25.4	26.0	25.5	0.5	3013	3.5
120	29.2	30.0	30.4	29.9	0.6	6026	7.9
180	33.4	34.4	34.6	34.1	0.6	9039	12.1

t (s)	Trial 1 (°C)	Trial 2 (°C)	Trial 3 (°C)	Mean *T* (°C)	± ½ range (°C)	*Q* (J)	Δ*T* (°C)
240	37.6	38.6	38.9	38.4	0.7	12052	16.4
300	41.6	42.9	43.0	42.5	0.7	15065	20.5
360	45.7	47.0	47.3	46.7	0.8	18077	24.7
420	49.6	51.2	51.3	50.7	0.9	21090	28.7
480	53.5	55.1	55.2	54.6	0.9	24103	32.6

Note: rounding ± ½ range to 1 d.p.; values below ±0.5 °C would be replaced by the thermometer's instrument uncertainty.

Typical trendline (excluding t = 0 point)

- $\Delta T = 1.38 \times 10^{-3}\, Q - 0.45$ $R^2 = 0.9997$
- Including all points: gradient 1.37×10^{-3} °C J^{-1}, intercept –0.28 °C, $R^2 = 0.9997$.
- $c = 1 / (0.150 \times 1.38 \times 10^{-3}) \approx 4.8 \times 10^3$ J kg^{-1} K^{-1}; percentage error ≈ +15%.
- Class results typically fall between about 4300 and 5500 J kg^{-1} K^{-1} – almost always **above** 4180 J kg^{-1} K^{-1}. Values below 4180 usually indicate a calculation error (e.g. mass in grams, or thermometer touching the element).
- The ideal gradient for 0.150 kg is $1/(0.150 \times 4180) = 1.59 \times 10^{-3}$ °C J^{-1}. The data from 60 s onward is very linear; the first minute shows a small lag.

Gradient interpretation

Rearranging $Q = mc\Delta T$ with ΔT as the subject:

$$\mathbf{\Delta T = (1 / mc) \times Q + 0}$$

y = mx + c	Rearranged equation
y	Δ*T*
x	*Q*
gradient	1/(mc)
y-intercept	0 (theoretical)

- **Gradient = 1/(mc)**, units °C J^{-1} (≡ K J^{-1}). Units check: 1/(kg × J kg^{-1} K^{-1}) = K J^{-1} ✓
- Therefore **c = 1 / (m × gradient)**, with m in kg.
- **Theoretical y-intercept = 0**: with no heat added there is no temperature change, so the line should pass through the origin (direct proportionality).
- **Actual intercept** is usually small and **negative** (about –0.3 to –1 °C). Physically, the x-intercept (≈ 200–400 J here) is the heat absorbed by the heater, cup, thermometer and lid before the water temperature begins to rise steadily – a "lag". A **positive** intercept usually means the thermometer was close to/touching the element, or the water was poorly stirred early on.
- The worksheet deliberately does not state what the gradient equals. Students who write "gradient = *c*" have skipped the rearrangement – send them back to part (c).

Model answers – evaluation

Q14 Reliability

- (a) The spread is smallest at the start and grows during heating (e.g. ±0.5 °C at 0–60 s up to ±0.9 °C at 420–480 s). Differences in starting temperature, stirring rate and the heater still being warm from a previous trial accumulate over time, and small timing differences matter more when temperature rises quickly.
- (b) $R^2 \approx 0.999$ means almost all of the variation in ΔT is explained by the linear relationship with Q; the mean points show very little random scatter about the trendline, so the data are precise/reliable. Note that a high R^2 says nothing about accuracy – a systematic error shifts the whole line without increasing scatter.
- (c) Random errors: parallax and interpolation when reading the thermometer (±0.5 °C); reaction time in reading temperature exactly at each 60 s mark; uneven stirring causing temperature gradients (hot water near the element, cooler water near the thermometer); fluctuations in current/voltage. Each causes readings to be randomly above or below the true value, producing the spread between trials.

Q15 Validity

- (a) Heat is absorbed by the heater casing, foam cup, lid and thermometer; heat is lost to the surroundings by conduction through the cup, convection/evaporation from the surface and through the lid holes; a small amount of energy is dissipated in the connecting leads.
- (b) Because some of Q does not go into the water, the measured ΔT is smaller than it would be for the same Q if all the energy went into the water. Losses increase as the water gets hotter (larger temperature difference to the room), so later points fall further below the ideal line.
- (c) **Too high.** $c = Q/(m\Delta T)$. Q is calculated from VIt, so it includes the energy lost; ΔT is smaller than it should be. A smaller denominator gives a larger c. Equivalently, the gradient $\Delta T/Q = 1/(mc)$ is smaller than ideal, so $c = 1/(m \times \text{gradient})$ is larger. The typical +15% error supports this.
- (d) Thermometer position (near the element reads too high; near the wall too low); poor mixing; V and I drifting as the element heats (resistance change), so $Q = VIt$ using a single reading is not exact; heater not fully submerged for the full trial.

Q16 Classification

Limitation	Type	Mainly affects
Heat lost to surroundings / absorbed by cup and heater	Systematic	Validity
Evaporation from water surface	Systematic	Validity
Thermometer reading (parallax, interpolation)	Random	Reliability
Reaction time reading temperature at 60 s marks	Random	Reliability
Uneven stirring / temperature gradients	Random (systematic if thermometer is always close to the element)	Reliability (validity)
Heater still warm from previous trial	Random between trials	Reliability

Q17 Improvements (model examples)

The change suggested	The problem it addresses	How the change addresses it
Use a double-walled insulated cup (or cup inside a larger foam cup) with a tight-fitting lid, sealing the lid holes with cotton wool.	Heat lost to surroundings (systematic; makes c too high).	Better insulation reduces conduction and evaporation losses, so a larger fraction of VIt goes into the water and Δ*T* is closer to its true value.
Start with water about 5–10 °C below room temperature and heat to about the same amount above room temperature.	Heat exchange with surroundings.	Heat gained from the room while below room temperature roughly cancels heat lost while above it, reducing the net systematic error.
Use a digital temperature probe with a data-logger recording automatically every 60 s.	Reading/parallax and reaction-time errors (random).	Removes human reading and timing error; resolution ±0.1 °C reduces uncertainty and spread between trials.
Cool the heater in cold water for 2 minutes and use water from the same container (same starting temperature) for every trial.	Different starting conditions between trials (random).	All trials start from the same state, reducing the spread between trials and improving reliability.
Use a joulemeter, or log V and I every 60 s and use the average power.	V and I drifting during heating.	Gives a more accurate measure of the actual energy supplied, so Q values are valid.

Q18 Model conclusion

The graph of ΔT against Q was linear (R^2 = 0.9997) with a y-intercept close to zero (–0.45 °C, within the ±0.5 °C uncertainty of the data), so the temperature change of 150 g of water is directly proportional to the heat added, supporting the aim. From the gradient (1.38×10^{-3} °C J^{-1}), the specific heat capacity of water was calculated as 4.8×10^3 J kg^{-1} K^{-1}, 15% higher than the accepted value of 4180 J kg^{-1} K^{-1}. The results were reliable (small spread between trials, high R^2), but the validity was limited by heat losses to the cup and surroundings, a systematic error that made the experimental value of c too high.

Common student misconceptions

- **Gradient = c.** Students read “the gradient is a property of water” and write gradient = c without rearranging. The gradient is 1/(mc); it also depends on the mass.
- **Heat and temperature are the same thing.** Heat (J) is energy transferred; temperature (°C, K) measures the average kinetic energy of particles. Equal heat gives different Δ*T* for different masses.
- **Using temperature instead of temperature change** on the y-axis, giving an intercept equal to room temperature and a confusing interpretation.
- **Mass in grams.** Using m = 150 instead of 0.150 kg gives c about 1000 times too small.
- **Converting ΔT to kelvin by adding 273.** A temperature change in °C is numerically equal to the change in K – no conversion is needed for Δ*T*.
- **“Heat loss makes c too low.”** Students reason that “less energy means lower c”. Walk through c = Q/(mΔT): Q (calculated) is unchanged, ΔT is smaller, so c is larger.
- **High R^2 means accurate.** R^2 measures how well the points fit a line (precision/reliability), not whether the line is correct. Systematic heat loss gives $R^2 \approx 1$ but c ≈ 15% high.
- **More trials fix systematic error.** Repeating reduces the effect of random error only; systematic losses are present in every trial.
- **“Human error” as a limitation.** Push students to name the specific error (e.g. parallax when reading the thermometer, reaction time at the 60 s mark) and its effect.

- **Forcing the trendline through the origin** to “make it proportional” – this hides the intercept they are asked to interpret.
- **Ammeter/voltmeter placement.** Voltmeter in series or ammeter in parallel (the latter can blow the ammeter fuse) – check circuits before switching on.